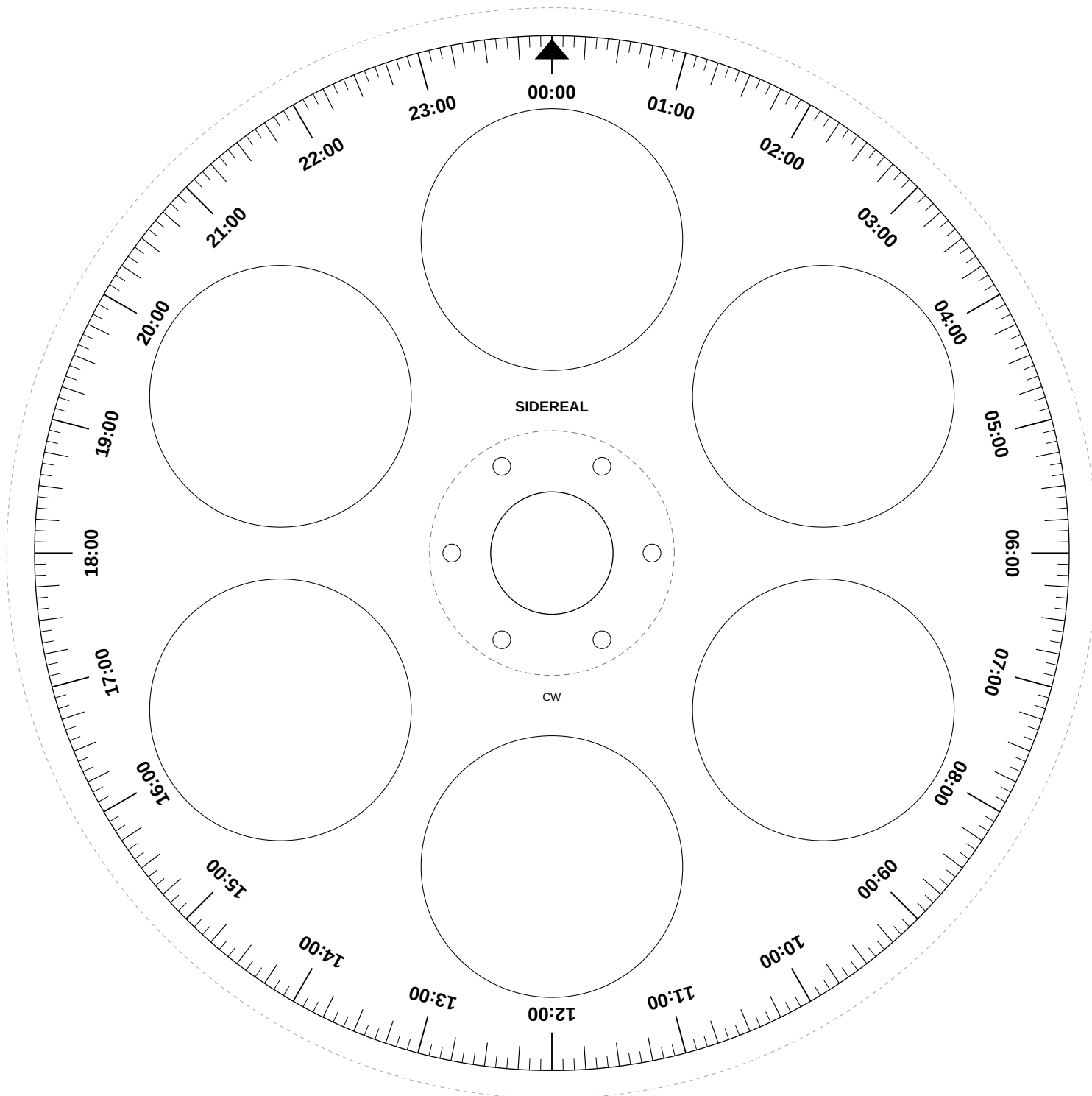




24 h SIDEREAL DIAL - final gear, 96 T module 2.0

Direction: HH increases CLOCKWISE viewed from above (printed side up).

One revolution = one sidereal day = 86164.1 s. 01:00 = one sidereal hour = 59 m 50.17 s

of civil time - this reads hour angle, not wall clock.

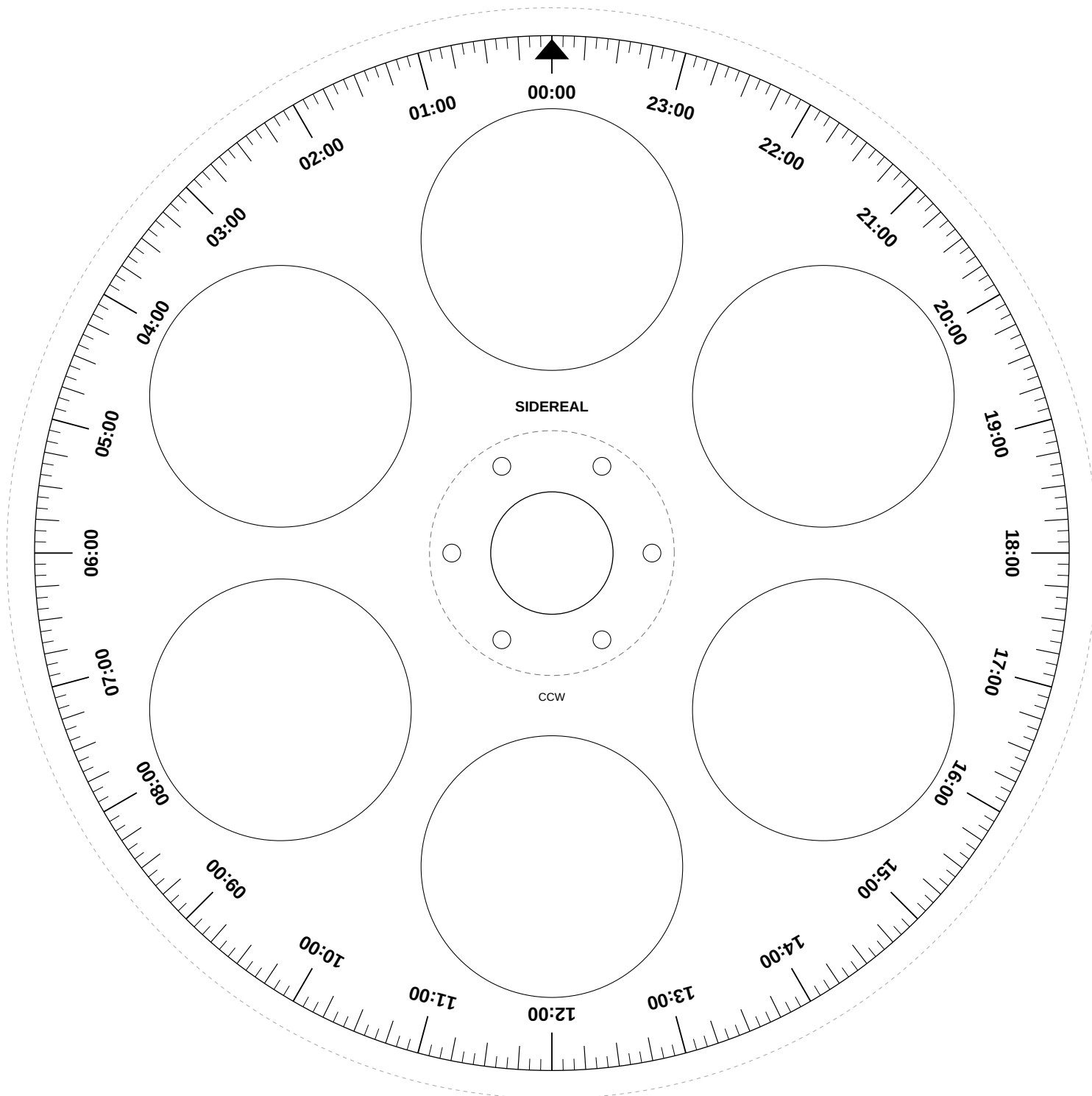
1 tooth = 3.75 deg = 897.5 s = 14 m 57.5 s. 96 teeth over 24 h, so every 15-minute tick is exactly one tooth pitch - use them to clock the dial.

Cut outs (build.ps1: bore=22, bc_holes=6, bc_r=18): outer 186.0 dia (0.5 inside the 187.0 root), 22 dia register bore, 6 x 3.2 dia bolts on r 18, 6 x 47.034 dia lightening holes on r 56.35 at 30 deg + i*60 deg. Those reach r 79.87, so all graduation sits in the solid band r 79.87 to 93.5. Outer dashed = tooth tips (196 dia); inner dashed 44 dia = clamp-hub footprint - trim there and skip the bolts.

PRINT AT 100% (page scaling off) - verify with the bar below.



measure this: must be exactly 100 mm



24 h SIDEREAL DIAL - final gear, 96 T module 2.0

Direction: HH increases COUNTER-CLOCKWISE viewed from above (printed side up).
 One revolution = one sidereal day = 86164.1 s. 01:00 = one sidereal hour = 59 m 50.17 s
 of civil time - this reads hour angle, not wall clock.
 1 tooth = 3.75 deg = 897.5 s = 14 m 57.5 s. 96 teeth over 24 h, so every 15-minute
 tick is exactly one tooth pitch - use them to clock the dial.

Cut outs (build.ps1: bore=22, bc_holes=6, bc_r=18): outer 186.0 dia (0.5 inside the
 187.0 root), 22 dia register bore, 6 x 3.2 dia bolts on r 18, 6 x 47.034 dia
 lightening holes on r 56.35 at 30 deg + i*60 deg. Those reach r 79.87, so all
 graduation sits in the solid band r 79.87 to 93.5. Outer dashed = tooth tips (196
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